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Introduction to testing 1m



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What is testing?

Testing is a process that happens during each of the planning, preparation and review stages of developing a web application. It's done to make sure the web application meets the set requirements.

Testing shows that the web application works properly and serves its intended purpose. It also helps to find any problems or faults in the application.

The purpose of testing is to ensure that all aspects of a web application function properly. This includes making sure that the web application is usable, has good performance and security and that there are no bugs.

Why do you need to do testing?

Testing provides information about the web application, which can increase an organisation or client's confidence in their web application or enable them to find defects (bugs) and correct them as necessary.

Testing builds quality into an application through its thorough processes. Testing aims to deliver error-free applications.

It's not possible to test every permutation or situation, so it's important to understand:

- why you need to test something
- what you are going to test
- how you are going to test it.

There are two key elements to testing.

- **Verification:** This checks that the web application is being developed according to the requirements and agreed design specifications. That is, it's checking precision and compliance.
- **Validation:** This evaluates the web application to determine if it meets the customer expectations and needs. That is, it's checking the value of a product for the

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Verification and validation are abbreviated as 'V&V' in some literature.

At first, these can seem very similar; both are critical components of quality management (also part of the ISO9000 and ISO9001 standards), so it's important to understand each and how they differ. Check your understanding by answering the following question.



☒ Verification vs validation

Non-Assessed

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Read the following statements and then match them to either verification or validation testing.

Verification	Validation
⌘ Determines if the web application you have built is correct and complies with specifications. ⌘ Testing occurs at every stage of development. ⌘ Determines if you have built a web application that meets the client's and customer's needs. ⌘ Testing occurs at the end of development, once the web application has been built.	

Submit



Seven principles of testing

Because it is not possible to test every possible permutation or combination, you need a good testing strategy to minimise the time and cost while still achieving your testing goals. This is done by applying good test principles and often by automating the testing.

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We will learn about automated testing later in this topic.

There are seven principles of testing that are used to reduce the possibility of undiscovered defects.

Watch *What are the 7 testing principles?* (Software Testing 101, 2022) (9:31) to learn about each of the seven principles.

Select Play to begin.



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What are the 7 Testing principles?

What are the 7 Testing principles?

Software Testing 101



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Principles of testing

Non-Assessed

Suppose you have 15 input fields to test, each having five possible values. The number of combinations to be tested would be 5^{15} .

How would you tackle this situation?

B <i>I</i> <u>U</u>      T ² T ₂ 
0 / 100 Word Limit

Save draft Submit



Software quality assurance

The *Guide to the Software Engineering Body of Knowledge (SWEBOK Guide)* (Washizaki, 2025), written by the IEEE Computer Society, is a book that brings together the knowledge, skills, techniques, methods, best practices and experiences developed in the field of software engineering over time. It's also recognised as an ISO standard, specifically ISO/IEC TR 19759.

Chapter 5 of the SWEBOK guide, *Software Testing Fundamentals* (Washizaki, 2025), offers a solid introduction to software testing within the field of software engineering. This chapter, which is about 40 pages long, is highly recommended if you're seriously considering software testing as a career path. For this course, it's a helpful read because it provides a broad overview of testing, whereas this topic focuses more specifically on testing in web development.



Thoughts on this page?

